

Hyungseok Ryu

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EDUCATION

2024 - present PhD Subject at **Gwangju Institute of Science and Technology**
2022 - 2023 Master's Degree at **Gwangju Institute of Science and Technology**
2016 - 2021 Bachelor's Degree at **Pusan National University**

RESEARCH EXPERIENCE

System Development for Humanoid Robot Locomotion Mar 2025 – present
End-to-end development of a humanoid robot system, including sensor integration, actuator control, communication architecture, and locomotion strategy. Reinforcement learning is applied to train robust gait policies in simulation for future deployment to physical hardware.

Improving Performance of Wearable Lower Limb Assistive Robots Mar 2023 - present
Using Free Energy Principle to enhance the performance of assistive robots. The research focuses on optimizing the interaction between the device and the user for improved mobility.

Estimating Gait Asymmetry and Gait Speed Jan 2023 - Present
Developing methodologies to estimate gait asymmetry and speed using sensor data embedded in wearable gait assistive robots, aiming to apply these insights in real-world settings for individuals with mobility impairments.

Recurve Archer Performance Improvement Jun 2023 - Jun 2025
Employed IMU sensors on upper limbs of recurve archers to collect shooting motion data. Analyzed this data to assess and predict the athletes' performance, aiming to provide insights for training enhancements.

Walking Simulation using OpenSim Oct 2022 - Jun 2023
Created and analyzed walking simulations for individuals with powered transfemoral prostheses using OpenSim, incorporating motion capture and force plate data. The findings were presented at the 2023 IEEE-ROMAN conference.

PUBLICATIONS

류형석, 홍우림, 이진원, et al. (2022). “보행속도에 따른 보행위상추정에 몸통 운동학의 영향”. In: *대한기계학회 춘추학술대회*, pp. 73–73.

Ryu, Hyungseok et al. (2023). “Towards Realistic Prosthetic Gait Simulations: Enhancing the Accuracy of OpenSim Analysis by Integrating the Transfemoral Prosthesis Model”. In: *2023 32nd IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*. IEEE, pp. 600–606.

류형석 and 허필원 (2023). “환측의 IMU와 보행시점 정보를 활용한 보행 비대칭성과 속도 추정”. In: *2023 대한생체역학회 공동학술대회*. 대한생체역학회.

류형석 et al. (2023). “현실적인 의족 보행 시뮬레이션을 향해: 대퇴의족 모델을 통합하여 OpenSim 분석의 정확도 향상하기”. In: *대한기계학회 춘추학술대회*, pp. 34–34.

류형석 and 허필원 (2025). “능동 추론 기반 제어 및 외란 토크 추정”. In: *2025 제어로봇시스템학회*.

AWARDS

Best Paper Award 2025 ICROS - "능동 추론 기반 제어 및 외란 토크 추정"

SKILLS

Technical Skills	Programming (Python, C++, MATLAB), Biomechanical Simulation and Analysis (OpenSim, Visual 3D), , Data Analysis
Language Proficiency	Korean (Native), English (Conversational)